DATASCI 226 Final

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```
library(readxl)
library(dplyr)
library(brms)
library(bayesplot)
library(ggplot2)
library(tidyr)
library(knitr)

set.seed(226)
```

1 Scientific Question

Do early changes in tumor volume predict pathologic complete response (pCR)?

2 Data Preparation

```
clin <- read_excel("data/ISPY2-Imaging-Cohort-1-Clinical-Data.xlsx")
mri  <- read_excel("data/Multi-feature-MRI-NACT-Data.xlsx")

df <- merge(clin, mri,
            by.x = "Patient_ID",
            by.y = "CLINICAL-TRIAL-SUBJECT-ID") |>
  select(pCR, FTV_pch_T0_T1, VOLUME_TUM_BLU_V10) |>
  filter(complete.cases(pick(everything()))) |>
  mutate(
    pCR       = as.integer(pCR),
    ftv_early = scale(FTV_pch_T0_T1)[,1],
    vol_base  = scale(VOLUME_TUM_BLU_V10)[,1]
  )

cat("N =", nrow(df), "| pCR rate =", round(mean(df$pCR), 3))
```

```
N = 384 | pCR rate = 0.294
```

Predictors:

- `FTV_pch_T0_T1`: % change in functional tumor volume from T0 to T1 (scaled). More negative = greater early shrinkage.
- `VOLUME_TUM_BLU_V10`: baseline tumor volume at T0 in cm^3 (scaled).

3 Model Specification

3.1 Primary Model (M1): Early FTV change + baseline volume

$$\text{logit}(P(\text{pCR}_i = 1)) = \beta_0 + \beta_1 \cdot \text{ftv_early}_i + \beta_2 \cdot \text{vol_base}_i$$

```r
priors_m1 <- c(
  prior(normal(0, 1.5),    class = Intercept),
  prior(normal(-0.8, 0.5), class = b, coef = ftv_early),
  prior(normal(-0.5, 0.5), class = b, coef = vol_base)
)
```

```r
m1 <- brm(
  pCR ~ ftv_early + vol_base,
  data   = df,
  family = bernoulli(link = "logit"),
  prior  = priors_m1,
  chains = 4,
  iter   = 4000,
  warmup = 1000,
  seed   = 226,
  file   = "output/m1"
)
```

3.2 Comparison Model (M2): Baseline volume only

M2 excludes early FTV change to test whether the imaging response adds predictive value beyond baseline tumor size alone.

```r
m2 <- brm(
  pCR ~ vol_base,
  data   = df,
  family = bernoulli(link = "logit"),
  prior  = c(
    prior(normal(0, 1.5),    class = Intercept),
    prior(normal(-0.5, 0.5), class = b, coef = vol_base)
  ),
```

```
  chains = 4,
  iter   = 4000,
  warmup = 1000,
  seed   = 226,
  file   = "output/m2"
)
```

4 Priors

Predictors are standardized (mean 0, SD 1).

Parameter	Prior	Justification
β_0 (Intercept)	$\mathcal{N}(0,\ 1.5)$	Weakly informative; consistent with ~29% observed pCR rate
β_1 (Early FTV change)	$\mathcal{N}(-0.8,\ 0.5)$	Early FTV reduction is an established predictor of pCR in NACT trials; negative direction well-supported
β_2 (Baseline volume)	$\mathcal{N}(-0.5,\ 0.5)$	Larger baseline tumors are less likely to achieve complete response; moderate negative effect expected

Both continuous predictors have informative priors based on published neoadjuvant chemotherapy evidence.

5 Posterior Results

```
coef_tbl <- fixef(m1, probs = c(0.025, 0.975)) |>
  as.data.frame() |>
  round(3)

combined_tbl <- data.frame(
  Estimate   = coef_tbl$Estimate,
  `2.5%`     = coef_tbl$Q2.5,
  `97.5%`    = coef_tbl$Q97.5,
  OR         = round(exp(coef_tbl$Estimate), 3),
  `OR 2.5%`  = round(exp(coef_tbl$Q2.5), 3),
  `OR 97.5%` = round(exp(coef_tbl$Q97.5), 3),
  check.names = FALSE
)
rownames(combined_tbl) <- rownames(coef_tbl)

kable(combined_tbl, caption = "Posterior coefficients and odds ratios (95% CrI) - M1")
```

Table 2: Posterior coefficients and odds ratios (95% CrI) — M1

	Estimate	2.5%	97.5%	OR	OR 2.5%	OR 97.5%
Intercept	-1.049	-1.298	-0.805	0.350	0.273	0.447
ftv_early	-1.481	-2.032	-0.950	0.227	0.131	0.387
vol_base	-0.197	-0.486	0.061	0.821	0.615	1.063

```
mcmc_areas(m1,
           pars = c("b_Intercept", "b_ftv_early", "b_vol_base"),
           prob = 0.90) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Posterior distributions with 90% credible intervals") +
  theme_bw()
```

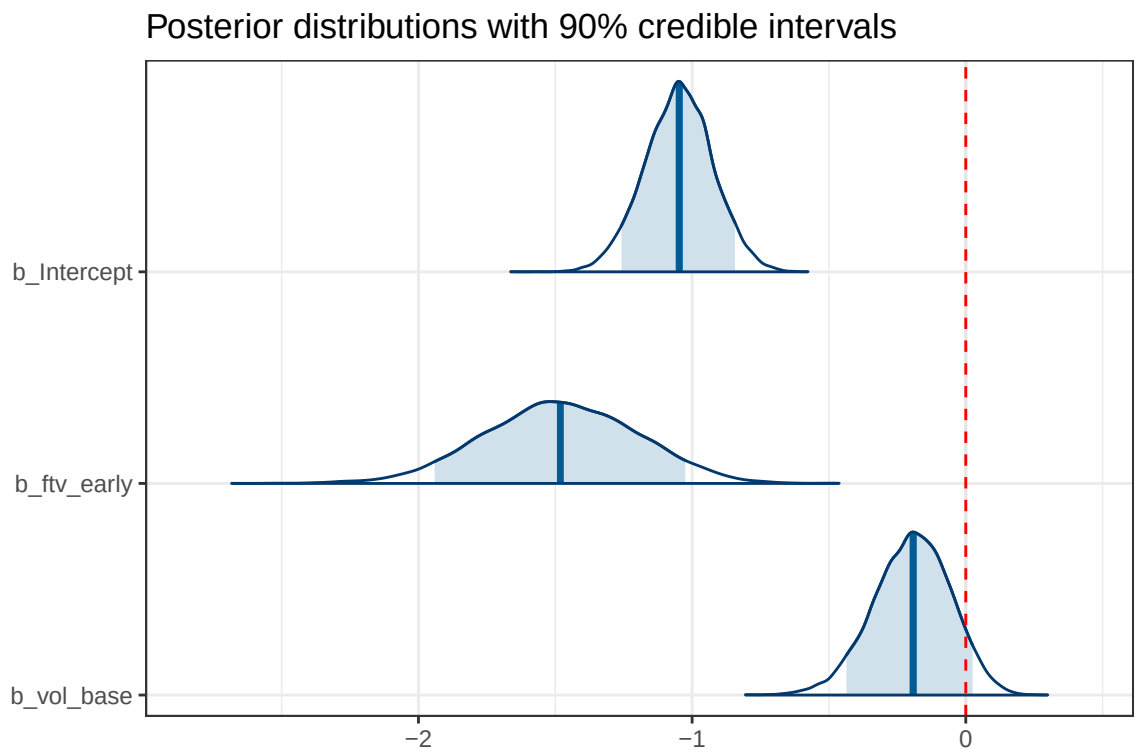

Figure 1: Posterior distributions of regression coefficients (M1)

Key findings:

- **Early FTV change** (β_1): Negative posterior — patients with greater early tumor shrinkage have higher odds of pCR.
- **Baseline volume** (β_2): Negative posterior — larger tumors at baseline associate with lower pCR odds, though with more uncertainty.

6 Model Checking

```
pp_check(m1, type = "bars", ndraws = 100) +
  labs(title = "Posterior predictive check (M1)",
       x = "pCR") +
  theme_bw()
```

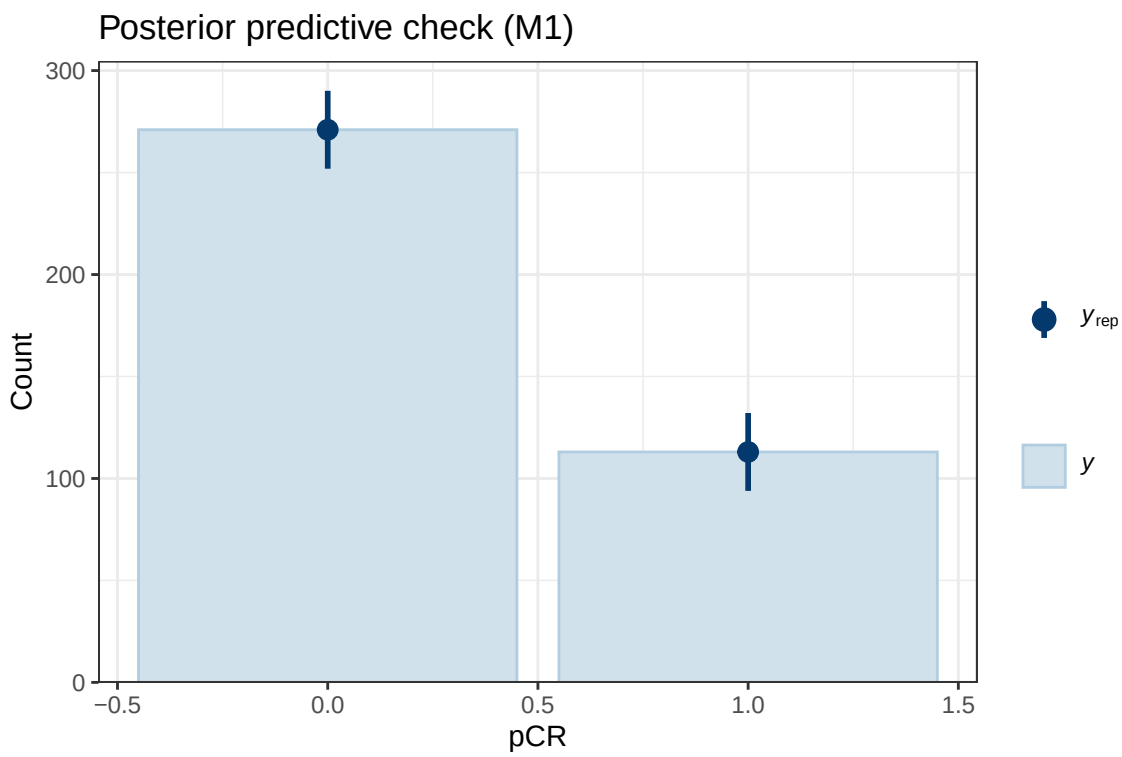

Figure 2: Posterior predictive check

Interpretation:

- Predicted 0/1 proportions closely match observed values.
- Posterior predictive means are centered on the observed rate (~0.29), indicating good calibration.

7 MCMC Diagnostics

```
m1_sum <- summary(m1)$fixed

kable(round(m1_sum, 3), caption = "MCMC diagnostics")
```

Table 3: MCMC diagnostics

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-1.049	0.125	-1.298	-0.805	1.000	8718.491	8524.177
ftv_early	-1.481	0.280	-2.032	-0.950	1.001	8128.146	8655.268
vol_base	-0.197	0.141	-0.486	0.061	1.000	10099.972	8240.997

- All $\hat{R} < 1.001$ — chains have converged ($\hat{R} \approx 1$ indicates mixing).
- Bulk ESS well above 1,000 for all parameters — posterior estimates are reliable.

```
mcmc_trace(m1,
           pars = c("b_Intercept", "b_ftv_early", "b_vol_base"),
           facet_args = list(ncol = 1)) +
  theme_bw()
```

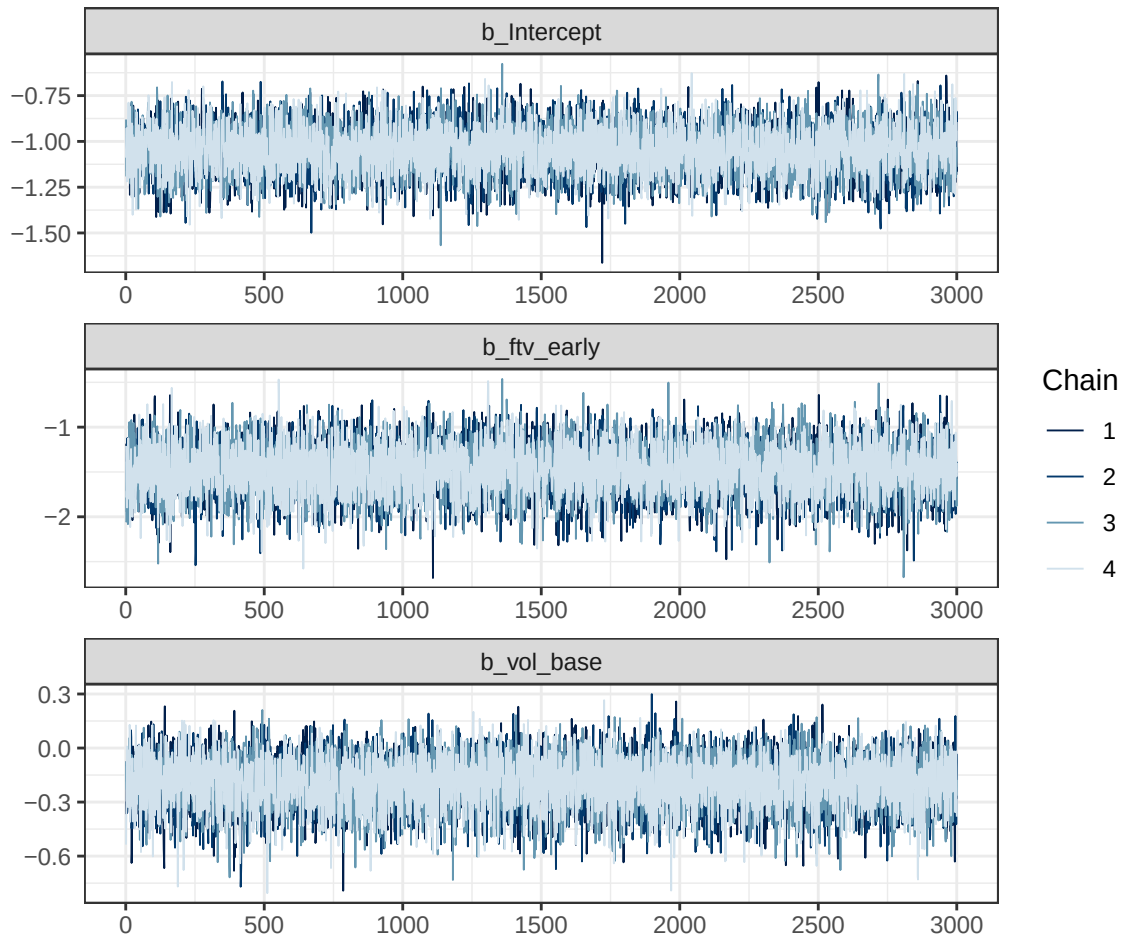

Figure 3: Trace plots for M1 regression coefficients

Trace plots show good chain mixing with no drift or divergence across all 4 chains.

8 Model Comparison

```
m1 <- add_criterion(m1, "loo")
m2 <- add_criterion(m2, "loo")

loo_comp <- loo_compare(m1, m2)
kable(as.data.frame(round(loo_comp, 2)),
      caption = "LOO-CV model comparison")
```

Table 4: LOO-CV model comparison

	elpd_diff	se_diff	elpd_loo	se_elpd_loo	p_loo	se_p_loo	looic	se_looic
m1	0.00	0.00	-218.31	8.74	2.62	0.36	436.63	17.49
m2	-15.76	5.06	-234.07	7.93	1.76	0.18	468.14	15.86

- If M1 is preferred, early FTV change carries predictive value beyond baseline tumor size alone.
- The ELPD difference relative to its standard error indicates whether the improvement is meaningful.

9 Sensitivity Analysis

```
m1_weak <- brm(
  pCR ~ ftv_early + vol_base,
  data   = df,
  family = bernoulli(link = "logit"),
  prior  = c(
    prior(normal(0, 10),  class = Intercept),
    prior(normal(0, 2.5), class = b)
  ),
  chains = 4,
  iter   = 4000,
  warmup = 1000,
  seed   = 226,
  file   = "output/m1_weak"
)
```

```
post_inf  <- as_draws_df(m1)[, c("b_ftv_early","b_vol_base")] |>
  mutate(model = "Informative")
post_weak <- as_draws_df(m1_weak)[, c("b_ftv_early","b_vol_base")] |>
```

```r
  mutate(model = "Weak")

bind_rows(post_inf, post_weak) |>
  pivot_longer(cols = starts_with("b_"), names_to = "parameter", values_to = "value") |>
  mutate(parameter = recode(parameter,
    b_ftv_early = "Early FTV change",
    b_vol_base  = "Baseline volume")) |>
  ggplot(aes(x = value, fill = model, color = model)) +
  geom_density(alpha = 0.4) +
  facet_wrap(~parameter, scales = "free") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(x = "Coefficient (log-odds)", y = "Density", fill = "Prior", color = "Prior") +
  theme_bw() + theme(legend.position = "bottom")
```

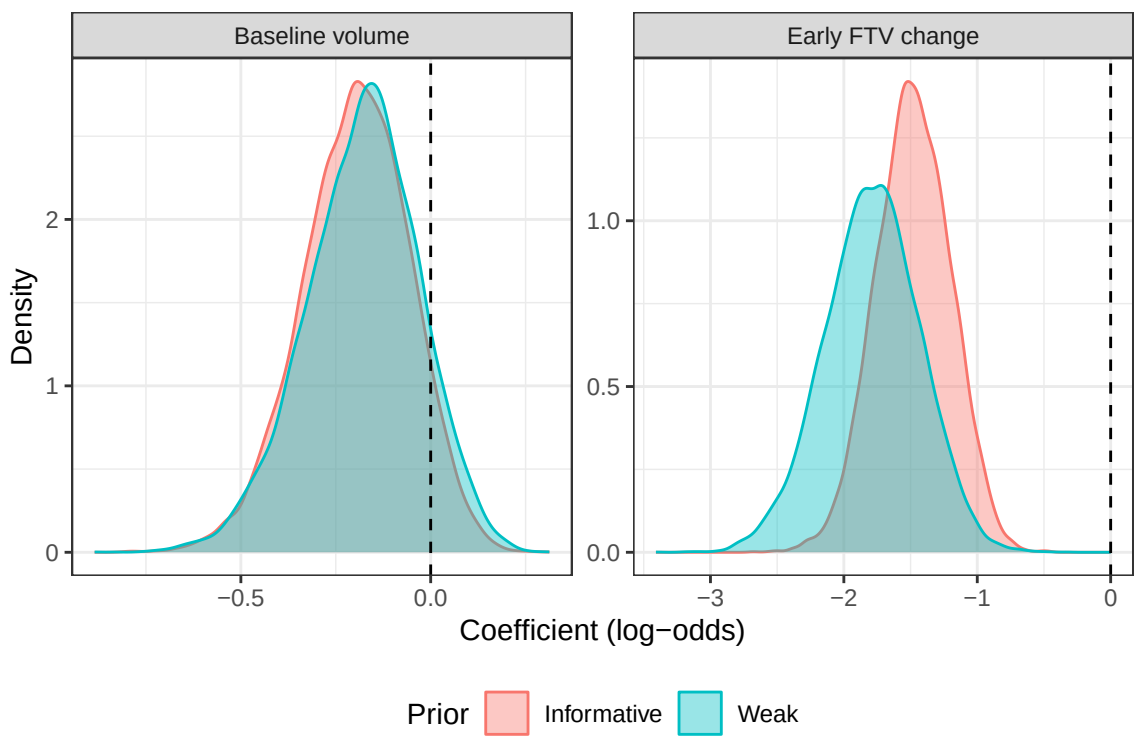

Figure 4: Posterior comparison: informative vs. weak priors

- Estimates are nearly identical under both prior specifications — the data drive the posterior.
- Conclusions are robust to prior choice.

10 Conclusions

- Greater early FTV reduction (T0→T1) is associated with higher probability of pCR, supporting early MRI response as a useful imaging biomarker.
- Baseline tumor volume shows a negative association with pCR but with more uncertainty.
- LOO-CV comparison confirms that early FTV change improves prediction over baseline size alone.
- Results are stable across informative and weak prior specifications.